Using dictionaries and analysing data

Lexicology and Lexicography — ***Course Website***

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Introduction

Recap

- Revisited why lexicology treats the lexicon as a **structured network**, not a flat list, using mental lexicon evidence and embedding demos.
- Distinguished **dictionary functions** (descriptive vs prescriptive), sense-ordering strategies, and usage labels that mediate interpretation.
- Practised translating OED search output into **research questions** on word classes, formation processes, and semantic domains.
- Walked through **OED workflows**: extracting post-2000 neologisms, exporting CSV data, and analysing them with tables and pivot charts.

Outline

1. Map the landscape of **dictionary types** (general, bilingual, specialised, diachronic vs synchronic).
2. Evaluate **crowdsourced vs curated** platforms (Urban Dictionary, Wiktionary, OED) and their affordances.
3. Explore **machine-readable resources** (WIND) and spreadsheet workflows for lexicological analysis.
4. Run a **hands-on comparison** of dictionary entries to surface researchable contrasts.
5. Connect dictionary outputs to **linguistic research applications** (historical, comparative, contemporary, methodological).

Types of Dictionaries

General Purpose Dictionaries

- Provide **comprehensive** linguistic information about words in a single language
  - → Spelling, pronunciation, and grammatical classification
  - → Multiple meanings and contextual usage examples
  - → Etymology and historical development notes
- Typically **monolingual** and organised alphabetically for easy reference
- Valuable for **morphological research** because they:
  - → Document detailed word origins and development
  - → Show common usage patterns and variations
  - → Provide historical context for word formation

Bilingual and Multilingual Dictionaries

- Facilitate **translation** between two or more languages
- Key features and functions:
  - → Support language learning and translation work
  - → Help decode meanings across different languages
  - → Illustrate cultural and linguistic nuances
- Important for morphological research by:
  - → Enabling **cross-linguistic** word formation studies
  - → Revealing patterns in **comparative linguistics**
  - → Showing how concepts transfer between languages

Specialized Dictionaries

- Focus on **specific aspects** of language or technical fields:
  - → Etymology and word origins
  - → Precise pronunciation guides
  - → Technical and discipline-specific terminology
  - → Slang and colloquial expressions
- Particularly valuable for:
  - → Detailed **academic research** in specific fields
  - → Understanding technical vocabulary development
  - → Tracing specialised word formation patterns

Diachronic and Synchronic Dictionaries

- **Diachronic** dictionaries:
  → Track how words change **through history**
  → Document evolution of meanings and forms
- **Synchronic** dictionaries:
  → Focus on language at **specific points in time**
  → Present contemporary usage and meanings
  → Document current word formation patterns
- Essential tools for:
  → **Historical linguistic research**
  → Understanding language evolution
  → Tracking word formation changes over time

Learner’s Dictionaries

- Specifically designed for **language acquisition**:
    - → Clear, simplified definitions
    - → Practical, everyday usage examples
    - → Word frequency information
    - → Common collocations and phrases
- Additional features often include:
    - → Grammar notes and usage guidelines
    - → Cultural context explanations
    - → Learning exercises and examples
- Useful for research into:
    - → Common **word usage patterns**
    - → Language acquisition processes
    - → Basic word formation principles

Specific Examples of Dictionaries

Oxford English Dictionary (OED)

- Official website: https://www.oed.com/
- Comprehensive **historical dictionary** of English by Oxford University Press
- Considered the **authoritative source** on English language
- Key features:
  - → **Historical approach**: traces word development chronologically
  - → **Extensive quotations**: over 3 million citations from wide-ranging sources
  - → **Comprehensive coverage**: 600,000+ words and phrases across 20 volumes
  - → Online since 2000; third edition in progress (electronic-only)
- Value for research:
  - → Traces detailed development of word forms and meanings
  - → Provides extensive historical context and usage patterns
  - → Enables in-depth **morphological analysis**
  - → Documents linguistic patterns and changes over time

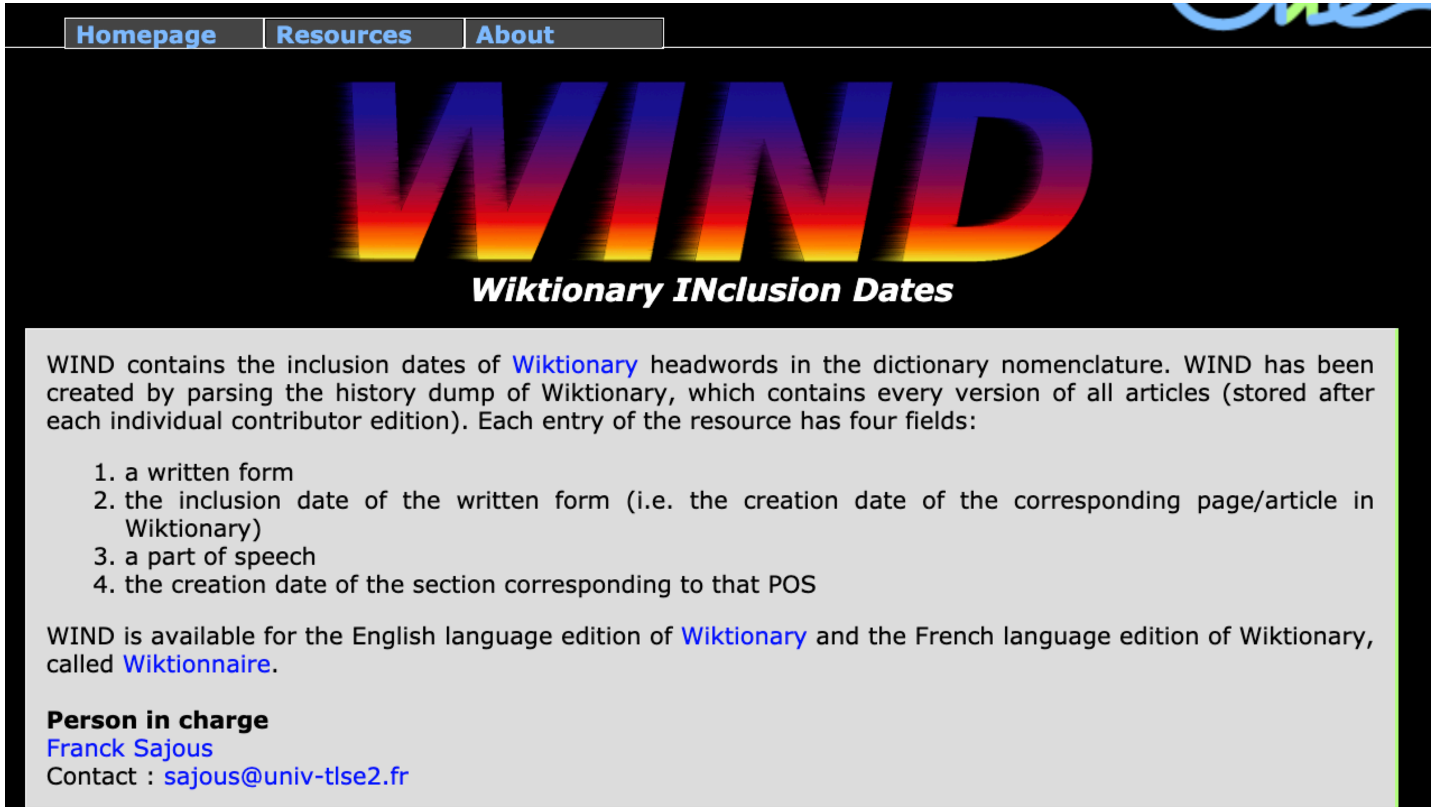
Wiktionary

- Official website: https://www.wiktionary.org/
- Multilingual, web-based **collaborative dictionary**
- Key features:
  - → **Crowdsourced content** with rapid updates
  - → Comprehensive and flexible entries for new words
  - → Multiple language support with cross-references
  - → Detailed etymological information where available
- Research applications:
  - → **Neologism** and language evolution studies
  - → **Cross-linguistic research** opportunities
  - → Contemporary usage patterns
  - → Educational tool for language learning and research

WIND (Machine-readable Wiktionary)

Sajous, Calderone, and Hathout (2020)

http://redac.univ-tlse2.fr/lexiques/wind.html



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WIND

Wiktionary INclusion Dates

WIND contains the inclusion dates of Wiktionary headwords in the dictionary nomenclature. WIND has been created by parsing the history dump of Wiktionary, which contains every version of all articles (stored after each individual contributor edition). Each entry of the resource has four fields:

1. a written form
2. the inclusion date of the written form (i.e. the creation date of the corresponding page/article in Wiktionary)
3. a part of speech
4. the creation date of the section corresponding to that POS

WIND is available for the English language edition of Wiktionary and the French language edition of Wiktionary, called Wiktionnaire.

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Excel Sheet

	A	B	C	D	E	F	G
1	Entry	PageId	Entry inclusion	#Revisions	POS	POS inclusion	#POS rev
2	enWiktHist_32.csv:"Loboda"	7339302	2019-09-01	4	NP	2019-09-01	1
3	enWiktHist_32.csv:"moving target"	7339196	2019-09-01	3	NC	2019-09-01	1
4	enWiktHist_32.csv:"baywing"	7339282	2019-09-01	2	NC	2019-09-01	1
5	enWiktHist_32.csv:"bombing run"	7339197	2019-09-01	2	NC	2019-09-01	1
6	enWiktHist_32.csv:"checkstand"	7339203	2019-09-01	2	NC	2019-09-01	1
7	enWiktHist_32.csv:"EPAC"	7339213	2019-09-01	2	NC	2019-09-01	1
8	enWiktHist_32.csv:"flight sequence"	7339198	2019-09-01	2	NC	2019-09-01	1
9	enWiktHist_32.csv:"none other than"	7339189	2019-09-01	2	ADJ	2019-09-01	1
10	enWiktHist_32.csv:"Shapoval"	7339277	2019-09-01	2	NP	2019-09-01	1
11	enWiktHist_32.csv:"slangery"	7339168	2019-09-01	2	NC	2019-09-01	1
12	enWiktHist_32.csv:"taraxasterol"	7339054	2019-09-01	2	NC	2019-09-01	1
13	enWiktHist_32.csv:"thiophosphoryl"	7339222	2019-09-01	2	NC	2019-09-01	1
14	enWiktHist_32.csv:"attb"	7339018	2019-09-01	1	NC	2019-09-01	1
15	enWiktHist_32.csv:"cash wrap"	7339204	2019-09-01	1	NC	2019-09-01	1
16	enWiktHist_32.csv:"döner kebab"	7339208	2019-09-01	1	NC	2019-09-01	1
17	enWiktHist_32.csv:"e-scooter"	7339284	2019-09-01	1	NC	2019-09-01	1
18	enWiktHist_32.csv:"go-ped"	7339220	2019-09-01	1	NC	2019-09-01	1
19	enWiktHist_32.csv:"landful"	7339011	2019-09-01	1	NC	2019-09-01	1
20	enWiktHist_32.csv:"lawnful"	7338994	2019-09-01	1	NC	2019-09-01	1
21	enWiktHist_32.csv:"libraryful"	7339012	2019-09-01	1	NC	2019-09-01	1
22	enWiktHist_32.csv:"ninety-ninety rule"	7339224	2019-09-01	1	NC	2019-09-01	1

Urban Dictionary

- Official website: https://www.urbandictionary.com/
- **Crowdsourced dictionary** focusing on **slang** and **colloquial language**
- Key features:
  - → **User-generated content** reflecting varied interpretations
  - → Dynamic tracking of language evolution
  - → Rich cultural context and references
  - → Multiple definitions showing usage variations
- Research value:
  - → **Sociolinguistic studies** of contemporary language
  - → Tracking emergence and spread of new slang
  - → Cultural analysis through language use
  - → Understanding informal **word formation processes**

Other Major Dictionaries

Merriam-Webster Dictionary (https://www.merriam-webster.com/)

- **American English** focus
- Strong etymology coverage
- Regular updates for contemporary usage
- Valuable for understanding American English word formation

Collins English Dictionary (https://dictionary.cambridge.org/)

- Comprehensive vocabulary coverage
- Strong **British English** focus
- Detailed word formation information
- Includes frequency information and usage trends

Cambridge English Dictionary (https://dictionary.cambridge.org/)

- Clear definitions and examples
- Strong **learner focus**
- British English emphasis
- Excellent grammatical information and usage notes

Online Etymology Dictionary (https://www.etymonline.com/)

- Detailed **word history** tracking
- **Etymology** focus
- Historical development emphasis
- Traces morphological changes over time

Applications in Linguistic Research

Historical linguistic analysis

- Tracking **morphological changes** through time
  → e.g. comparing Old English strong verbs to Modern English irregular forms
- Understanding **semantic development**
  → e.g. charting the shift of *awful* from 'full of awe' to 'terrible'
- Documenting **word formation patterns**
  → e.g. analyzing the emergence of the prefix *cyber*- in late 20th century

Comparative linguistics

- **Cross-linguistic** word formation studies
  → e.g. comparing German and English compound nouns (e.g. de. *Zahnarzt* vs en. *tooth doctor*)
- Analysis of **borrowing patterns**
  → e.g. tracing French loanwords (e.g. *cuisine*, *ballet*) into English after the Norman Conquest
- Understanding **morphological universals**
  → e.g. investigating prefixation productivity across unrelated languages (e.g. *ge*- in German, *en*- in English)

Contemporary language research

- **Neologism** studies and word creation
  → e.g. investigating the rise of *selfie* and *meme* via social media corpora
- **Sociolinguistic variation**
  → e.g. examining morphological variation in teenage text messaging (e.g. abbreviations like *u* for you)
- **Internet-influenced** language change
  → e.g. analyzing adoption of acronyms (e.g. *lol*, *brb*) in spoken and written registers

Methodological applications

- **Diachronic studies** of word formation
  → e.g. comparing *selfie* definitions in 2013 vs 2025 corpora to trace shifts from novelty to mainstream usage
- **Quantitative analysis** of lexical change over time
  → e.g. measuring the rise of *cryptocurrency* entries over time
- **Corpus-based validation** of dictionary entries
  → e.g. using the *Corpus of Contemporary American English* to test whether dictionary data match corpus data
- Tracking the adoption of **loanwords** and **neologisms**
  → e.g. tracing how *hygge* transitions from Danish loanword to standard entry in learner dictionaries

- Identifying gaps or biases in dictionary coverage
  - → e.g. noticing that everyday items like *takeaway* are absent from learner dictionaries
- Supporting **computational linguistics** and NLP tasks (e.g. lemmatisation, part-of-speech tagging)
  - → e.g. expanding sentiment lexicons with blended forms like *hangry*
- Informing the design of new dictionaries and lexical resources
  - → e.g. designing a climate change glossary after analysing coverage gaps for items like *carbon offsetting*

Practice: analysing and comparing dictionary entries

- Choose at least two terms that might be interesting for comparison.
- Look them up in ≥ 2 dictionaries from ≥ 2 different dictionary types.
- Analyse and compare the entries between the dictionaries.
- Can you think of related research questions?
- Prepare slides in this PowerPoint file.
- Present your findings to the class.

Practise using the OED

Analysis objectives

1. How many terms are marked as ***obsolete*** across the whole OED?
2. What are the most common **word classes** of obsolete terms?
3. What are the most common **word-formation processes**?
4. What are the most common **subjects**?

Using the OED's Advanced Search

- find all terms that are marked as ***obsolete***
- sort them by `Date (newest first)`
- export the results as a **CSV file**

Using Microsoft Excel

1. open the **CSV file**
2. save the file as an Excel file (`.xlsx` extension)
3. convert the exported rows/columns to a `Table`
4. create **`Pivot Table`s** to analyse the data
5. create **`Pivot Chart`s** to visualise the data

Model Excel file

Summary

- **Dictionaries are not monolithic.** Different types — general, specialised, bilingual, diachronic—serve different purposes.
- **Choose the right tool for the job.** The best dictionary for your research depends on your specific questions (e.g., historical development, contemporary slang, cross-linguistic comparison).
- **Dictionaries are data.** They are curated datasets that can be used for qualitative and quantitative linguistic analysis, revealing patterns in word formation, semantic change, and language use.

References

Sajous, Franck, Basilio Calderone, and Nabil Hathout. 2020. “ENGLAWI: From Human- to Machine-Readable Wiktionary.” In *Proceedings of the 12th Language Resources and Evaluation Conference*, 3016–26. Marseille, France: European Language Resources Association. https://www.aclweb.org/anthology/2020.lrec-1.369.